

Pricing Weather Derivatives Using Esscher Transforms

Utkarsh Sawarn | SURGE | Mentor: Prof. Sourav Majumdar

1. Introduction

Traditional crop insurance relies on manual damage assessments, often resulting in months-long settlement delays. Weather derivatives offer an alternative by triggering payouts automatically based on objective meteorological data, such as rainfall indices. This automation eliminates administrative bottlenecks, enabling faster capital allocation to affected agricultural sectors. Given the Indian economy's dependence on monsoon performance, efficient risk transfer mechanisms are essential. The introduction of indices like RAINMUMBAI demonstrates the growing viability of treating Indian weather risk as a tradable asset class.

Standard financial models, such as Black-Scholes, assume continuous, symmetric asset price trajectories. These models are inadequate for Indian monsoon data, which exhibits significant skewness, heavy tails, and discontinuous jump patterns. The increasing frequency of extreme weather events further limits the applicability of standard normal distributions for rainfall modeling.

2.Variance Gamma Processes

The Variance Gamma (VG) Process:

Monsoon precipitation is characterized by extended dry spells punctuated by intense rainfall. We represent this using a Variance Gamma (VG) Lévy Process, conceptualized as a Brownian motion evaluated along a random Gamma time change (G). This framework effectively models periods of inactivity interrupted by sudden jumps. Economically, extreme rainfall days accelerate the "financial clock" relative to calendar time, incorporating abrupt risk events into the pricing dynamic:

$$\phi_{VG}(u) = \left(1 - i\theta\nu u + \frac{1}{2}\sigma^2\nu u^2\right)^{-\frac{1}{\nu}}$$

Method of Moments Calibration:

We map variance (V), skewness (S), and kurtosis (K) using 25 years of daily precipitation data from the India Meteorological Department (IMD). This dataset encompasses multiple El Niño and La Niña cycles, incorporating extreme climate variance. The parameter ν governs the variance of the time change; higher ν values indicate clustered, severe rainfall events, whereas lower values represent more continuous precipitation. Boundary conditions are applied to ensure numerical stability:

- $\nu = \max\left(0.1, \frac{K-3}{3.0}\right)$ prevents negative time-change parameters during winter dry spells, maintaining stability when rainfall variance approaches zero.
- To prevent negative variance ($\sigma^2 \leq 0$) driven by high skewness, baseline volatility is constrained to 70% of the empirical standard deviation, with drift (θ) adjusted accordingly.

3. Payoff Structure & Market Simulation

The derivative utilizes a standard call option payoff structure. For simulated rainfall I_{sim} , strike level K , and tick size α (per mm), the payout is defined as:

$$\text{Payoff}(I_{sim}) = \alpha \times \max(I_{sim} - K, 0)$$

In the absence of an active Indian weather derivatives market, we estimate a target premium (P_{market}) under the physical probability measure ($\mathbb{P}$). We construct an Empirical Distribution Function (EDF) from historical data to proxy current market pricing. This EDF is applied alongside a 15% risk loading factor ($\lambda = 0.15$) and a 6.5% risk-free rate ($r = 0.065$):

$$P_{market}(K) = e^{-rT} \times \mathbb{E}^{\mathbb{P}}[\text{Payoff}] \times (1 + \lambda)$$

4. The Role Of Esscher Transforms

Weather derivatives operate in an incomplete market framework since precipitation is a non-tradable underlying asset. We apply the **Esscher Transform** to transition from the physical measure ($\mathbb{P}$) to a risk-neutral pricing measure ($\mathbb{Q}$), isolating the market risk-aversion parameter (h^*) that satisfies:

$$\mathbb{E}^{\mathbb{Q}}[\text{Payoff}](h^*) - e^{-rT} \mathbb{E}^{\mathbb{P}}[\text{Payoff}](1 + \lambda) = 0$$

Risk-Neutral Parameter Shifts:

Defining $\kappa(h) = 1 - \theta\nu h - 0.5\sigma^2\nu h^2$, the Esscher transform maps the physical parameters to their risk-neutral equivalents as follows:

$$\theta^{\mathbb{Q}} = \frac{\theta + \sigma^2 h}{\kappa(h)}, \quad \sigma^{\mathbb{Q}} = \frac{\sigma}{\sqrt{\kappa(h)}}, \quad \nu^{\mathbb{Q}} = \nu$$

This transformation shifts probability mass from expected empirical outcomes toward tail events, mathematically formalizing the risk premium required by underwriters to assume volatile weather exposures.

5. Results

Once derived for the baseline contract ($K = 0$ mm), h^* is treated as constant across all strike prices for the respective month.

Month	Mean (mm)	Theta (θ)	Sigma (σ)	Target Q	Final Premium
Jan	0.30	0.0651	0.8719	350.52	348.63
Mar	1.80	0.9193	4.7290	2,068.07	2,056.90
Jun	530.21	184.7814	223.9633	6,09,741.22	6,06,447.39
Jul	1024.49	-237.6392	373.2611	11,78,167.82	11,71,803.33
Aug	542.20	447.6806	216.9591	6,23,528.34	6,20,160.03
Oct	67.35	151.9946	47.1130	77,455.18	77,036.77

To calculate these final premiums, the calibrated h^* is applied to the moment generating function to derive risk-neutral parameters ($\theta^{\mathbb{Q}}, \sigma^{\mathbb{Q}}$). A Monte Carlo simulation of 50,000 iterations is then executed using these adjusted parameters; the resulting expected payouts are discounted to present value.

6. The Negative θ Paradox & Valuation Insights

The July Drift Phenomenon:

Calibration for July yields a significant negative drift parameter ($\theta = -237.6392$). In the VG framework, θ dictates skewness rather than strictly directional drift. This reflects the empirical characteristics of July precipitation: a high historical mean (~ 1024.49 mm) paired with a pronounced left tail representing acute intra-seasonal dry spells. This confirms the model's ability to accommodate asymmetric rainfall distributions.

Volatility-Driven Premium Scaling:

The calculated July premium (**11,71,803.33 Rs.**) is nearly double that of June, despite July possessing roughly twice the mean rainfall. This non-linear scaling is driven by July's substantial volatility ($\sigma = 373.2611$), which broadens the probability distribution. The pricing structure dynamically reflects the cost of this variance, appropriately scaling the required risk premium for wider tail exposures rather than relying solely on mean precipitation.

7. Social Imapct & Further Work

Boundary Enforcement:

Because continuous statistical processes can theoretically generate negative values, simulated rainfall is bounded using a non-linear $\max(\cdot, 0)$ function prior to payout calculation. This constraint prevents structural anomalies in the payout profile, ensuring that the financial modeling remains consistent with physical meteorological limits and preventing logical errors in automated settlement.

Systemic Agricultural Hedging:

A significant portion of the Indian workforce is exposed to monsoon variability, with delayed post-disaster relief often exacerbating rural debt cycles. This pricing framework provides a quantitative foundation for regional institutions to underwrite localized, over-the-counter weather derivatives. Farmer Producer Organizations (FPOs) can aggregate resources to purchase these contracts, effectively transferring climate risk to broader capital markets. The objective, automated payout mechanism ensures rapid liquidity in the event of rainfall deficits, allowing for immediate remediation strategies, such as replanting with drought-resistant crops. Ultimately, this scalable mathematical approach can be applied across various regional indices, standardizing climate risk management for the agricultural sector.

8. References

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